2.36 Volume of Revolution: Parametric Equations

Worked Solutions

1a)

When $x = 4$, $t = \sqrt{4} = 2$. Therefore

$$V = \pi \int_0^2 (t^3)^2 (2t)\, dt = \pi \int_0^2 2t^7\, dt = \frac{\pi}{4}[t^8]_0^2 = 64\pi \text{ units}^3.$$

1b)

When $x = 16$, $t = \sqrt[3]{8} = 2$. Therefore

$$V = \pi \int_0^2 (t^2)^2 (6t^2)\, dt = 6\pi \int_0^2 t^6\, dt = \frac{768\pi}{7} \text{ units}^3.$$

1c)

When $x = 28$, $t^3 + 1 = 28$, so $t = 3$. Therefore

$$V = \pi \int_0^3 (t^2)^2 (3t^2)\, dt = 3\pi \int_0^3 t^6\, dt = \frac{6561\pi}{7} \text{ units}^3.$$

1d)

$$V = \pi \int_0^{\pi/2} (\sqrt{\cos t})^2 \cos t\, dt = \pi \int_0^{\pi/2} \cos^2 t\, dt.$$

Using $\cos^2 t = \frac{1}{2}(1 + \cos 2t)$,

$$V = \pi \left[\frac{t}{2} + \frac{1}{4}\sin 2t\right]_0^{\pi/2} = \frac{\pi^2}{4} \text{ units}^3.$$

2a)

Since $y = 1 + t^2$, $dy/dt = 2t$ and $x^2 = 1 - t^2$. Thus

$$V = \pi \int_0^1 (1 - t^2)(2t)\, dt = \pi \left[t^2 - \frac{1}{2}t^4\right]_0^1 = \frac{\pi}{2} \text{ units}^3.$$

2b)

When $y = 1$, $\tan t = 1$, so $t = \pi/4$. Therefore

$$V = \pi \int_0^{\pi/4} (2\cos t)^2 \sec^2 t\, dt = 4\pi \int_0^{\pi/4} dt = \pi^2 \text{ units}^3.$$

2c)

Here $x^2 = \sin^4 t \cos t$ and $dy/dt = \sin t$. Therefore

$$V = \pi \int_0^{\pi/2} \sin^5 t \cos t\, dt = \frac{\pi}{6}[\sin^6 t]_0^{\pi/2} = \frac{\pi}{6} \text{ units}^3.$$

3)

When $x = e^2$, $t = 4$; when $x = e^3$, $t = 6$. Also $dx/dt = \frac{1}{2}e^{t/2}$ and $y^2 = t - 2$. Hence

$$\begin{aligned} V &= \pi \int_4^6 (t-2)\frac{1}{2}e^{t/2}\,dt \\ &= \frac{\pi}{2}\left[2e^{t/2}(t-2) - 4e^{t/2}\right]_4^6 \\ &= 2\pi e^3 \text{ units}^3. \end{aligned}$$

4)

For $1 \le t \le 3$, $x^2 = \ln(t+1)$ and $dy/dt = 2t$. Therefore

$$V = \pi \int_1^3 2t \ln(t+1)\,dt.$$

Integration by parts gives

$$\begin{aligned} V &= \pi \left[t^2 \ln(t+1) - \int \frac{t^2}{t+1}\,dt\right]_1^3 \\ &= \pi \left[t^2 \ln(t+1) - \frac{1}{2}t^2 + t - \ln(t+1)\right]_1^3 \\ &= 2\pi(8\ln 2 - 1) \text{ units}^3. \end{aligned}$$

5a)

When $y = 0$, $t = 1$; when $y = \ln 4$, $t = 4$. Since $x^2 = 16/t^2$ and $dy/dt = 1/t$,

$$V = \pi \int_1^4 \frac{16}{t^3}\,dt = \pi\left[-\frac{8}{t^2}\right]_1^4 = \frac{15\pi}{2} \text{ cm}^3.$$

5b)

The model assumes that the funnel has no thickness and perfectly follows the curve.

6a)

The curve meets the x-axis when $\sqrt{\sin t} = 0$. The non-zero endpoint in the stated interval is $t = \pi$, so $k = \pi$.

6b)

Since $dx/dt = \sin t$ and $y^2 = \sin t$,

$$\begin{aligned} V &= \pi \int_0^\pi \sin^2 t\,dt \\ &= \pi \int_0^\pi \left(\frac{1}{2} - \frac{1}{2}\cos 2t\right) dt = \frac{\pi^2}{2} \text{ units}^3. \end{aligned}$$

7)

Since $x^2 = t$ and $dy/dt = e^t$,

$$\begin{aligned} V &= \pi \int_0^k te^t\,dt = \pi[te^t - e^t]_0^k \\ &= \pi((k-1)e^k + 1). \end{aligned}$$

Equating this with $\pi(3e^4 + 1)$ gives $(k-1)e^k = 3e^4$, so $k = 4$.

8a)

Let $u = \sin\theta$, so $du = \cos\theta\, d\theta$. Since $\cos^3\theta = \cos\theta(1 - \sin^2\theta)$,

$$\int \cos^3\theta\, d\theta = \int (1 - u^2)\, du = u - \frac{1}{3}u^3 + c = \sin\theta - \frac{1}{3}\sin^3\theta + c.$$

8b)

Using the result in part (a),

$$\begin{aligned} V &= \pi \int_{-\pi/2}^{\pi/2} \cos^3\theta\, d\theta \\ &= \pi \left[\sin\theta - \frac{1}{3}\sin^3\theta\right]_{-\pi/2}^{\pi/2} = \frac{4\pi}{3} \text{ units}^3. \end{aligned}$$

9a)

Using $\cosh^2 t - \sinh^2 t = 1$,

$$xy = (\cosh t - \sinh t)(\cosh t + \sinh t) = 1,$$

so $y = 1/x$. For $t \le 0$, $x = e^{-t} \ge 1$.

9b)

$$V = \pi \int_1^\infty \frac{1}{x^2}\, dx = \pi \left[-\frac{1}{x}\right]_1^\infty = \pi \text{ units}^3.$$

10a)

At $t = 0$, $x = a = 2$ because the diameter is 4 cm. At $t = 2$, $y = 4b = 3$, so $b = 3/4$.

10b)

Using $x = 2e^{-t}$ and $dy/dt = \frac{3}{2}t$,

$$\begin{aligned} V &= \pi \int_0^2 (2e^{-t})^2 \frac{3}{2} t\, dt = 6\pi \int_0^2 te^{-2t}\, dt \\ &= 6\pi \left[-\frac{1}{2}te^{-2t} - \frac{1}{4}e^{-2t}\right]_0^2 \\ &= \frac{3\pi}{2}(1 - 5e^{-4}) \approx 4.28 \text{ cm}^3. \end{aligned}$$

10c)

The model assumes the nozzle has no thickness and that its inner surface exactly follows the curve.

11a)

The maximum value of $x = \sqrt{t^2 - 1}$ occurs at $t = 3$. Therefore the maximum diameter is

$$2\sqrt{3^2 - 1} = 4\sqrt{2} \approx 5.7 \text{ cm}$$

to the nearest millimetre.

11b)

The minimum height is $y = 1 + \ln 1 = 1$ at $t = 1$, and the maximum is $y = 3 + \ln 3$ at $t = 3$. The ornament's maximum height is therefore $2 + \ln 3 \approx 3.1$ cm to the nearest millimetre.

11c)

Since $x^2 = t^2 - 1$ and $dy/dt = 1 + 1/t$,

$$\begin{aligned} V &= \pi \int_1^3 (t^2 - 1)\left(1 + \frac{1}{t}\right) dt \\ &= \pi \left[\frac{1}{3}t^3 + \frac{1}{2}t^2 - t - \ln t\right]_1^3 \\ &\approx 30.1 \text{ cm}^3. \end{aligned}$$

11d)

Using the unrounded volume, the mass is approximately $2.5(30.1) = 75.25$ g, which is 75 g to two significant figures.

12a)

Differentiating $y = t^2 e^{-t}$ gives

$$\frac{dy}{dt} = te^{-t}(2 - t).$$

The maximum occurs at $t = 2$. Thus $x = 2/t = 1$ and $y = 4/e^2$, so $P = (1, 4/e^2)$.

12b)

When $x = 1$, $t = 2$; when $x = 3$, $t = 2/3$. Since $dx/dt = -2t^{-2}$ and $y^2 = t^4 e^{-2t}$,

$$\begin{aligned} V &= 2\pi \int_{2/3}^2 t^2 e^{-2t}\, dt \\ &= \pi \left[-\frac{1}{2}t^2 e^{-2t} - \frac{1}{2}te^{-2t} - \frac{1}{4}e^{-2t}\right]_{2/3}^2 \\ &= \pi \left(\frac{29}{18e^{4/3}} - \frac{13}{2e^4}\right). \end{aligned}$$

13)

When $x = \ln 13$, $t = 16$ and $y = 2$. When $x = 0$, $t = 4$. The line from the origin to P has equation $y = 2x/\ln 13$. The volume generated by the curve is

$$\begin{aligned} V_C &= \pi \int_4^{16} \left(\frac{8}{\sqrt{t}}\right)^2 \frac{1}{t - 3}\, dt \\ &= \pi \int_4^{16} \frac{64}{t(t - 3)}\, dt = \frac{64\pi}{3} \ln \frac{13}{4}. \end{aligned}$$

The volume generated by the straight line is

$$V_L = \pi \int_0^{\ln 13} \left(\frac{2x}{\ln 13}\right)^2 dx = \frac{4\pi \ln 13}{3}.$$

The region's volume is therefore

$$V_C - V_L = \frac{4\pi}{3}\left(16 \ln \frac{13}{4} - \ln 13\right) \text{ units}^3.$$